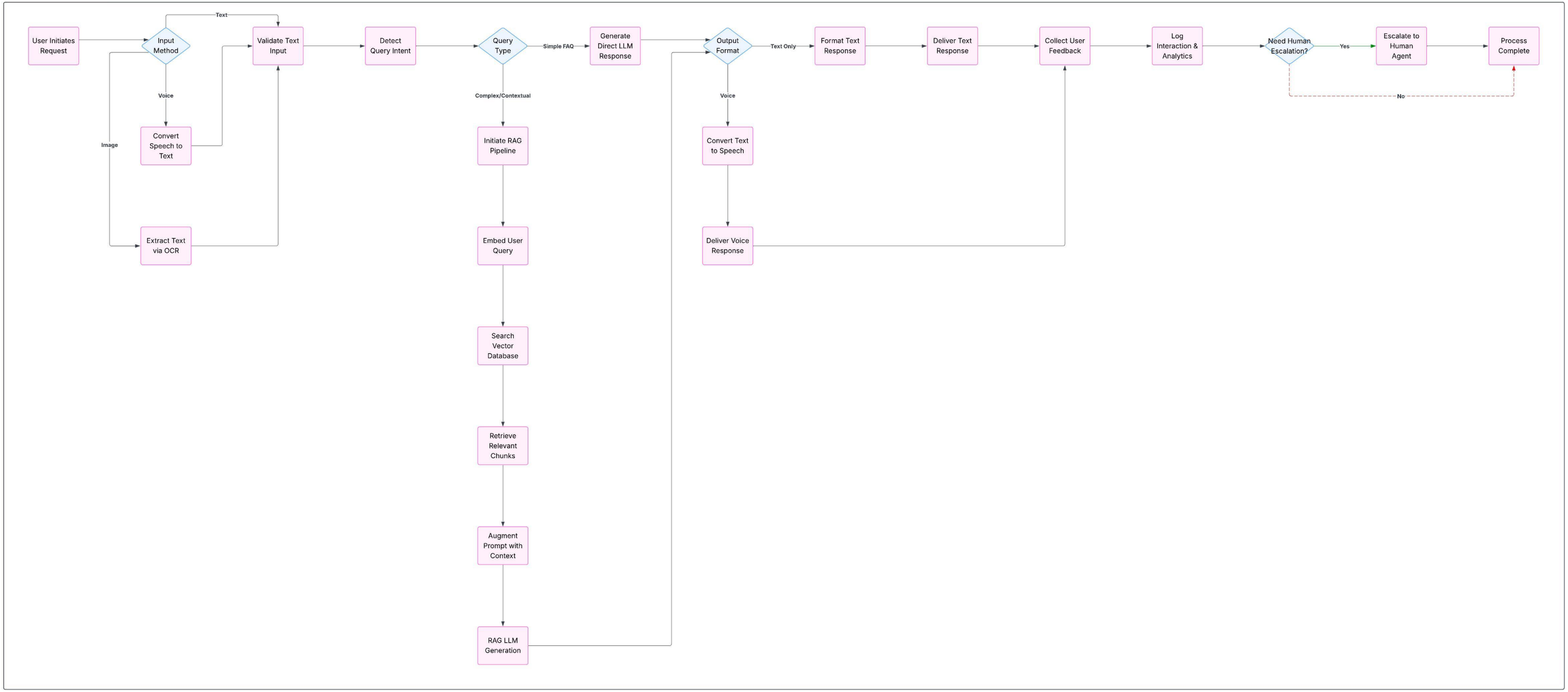
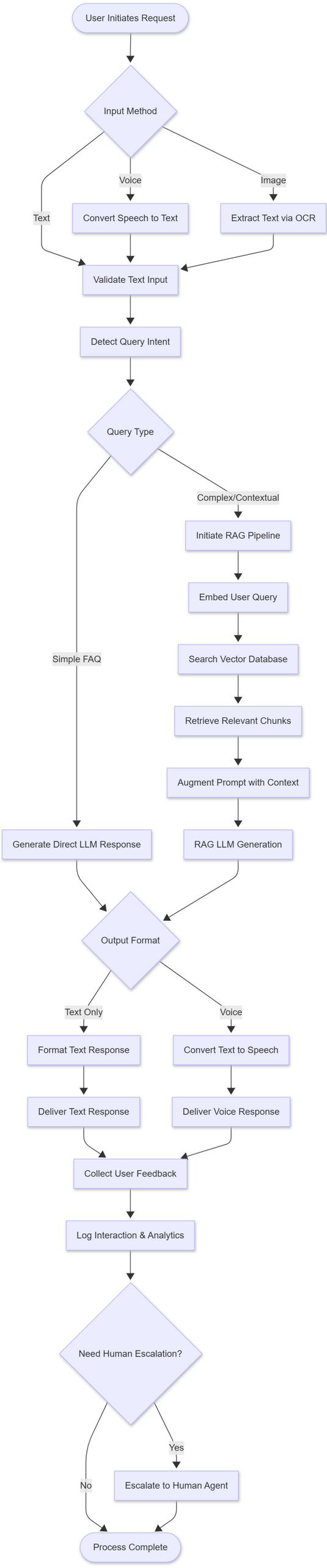
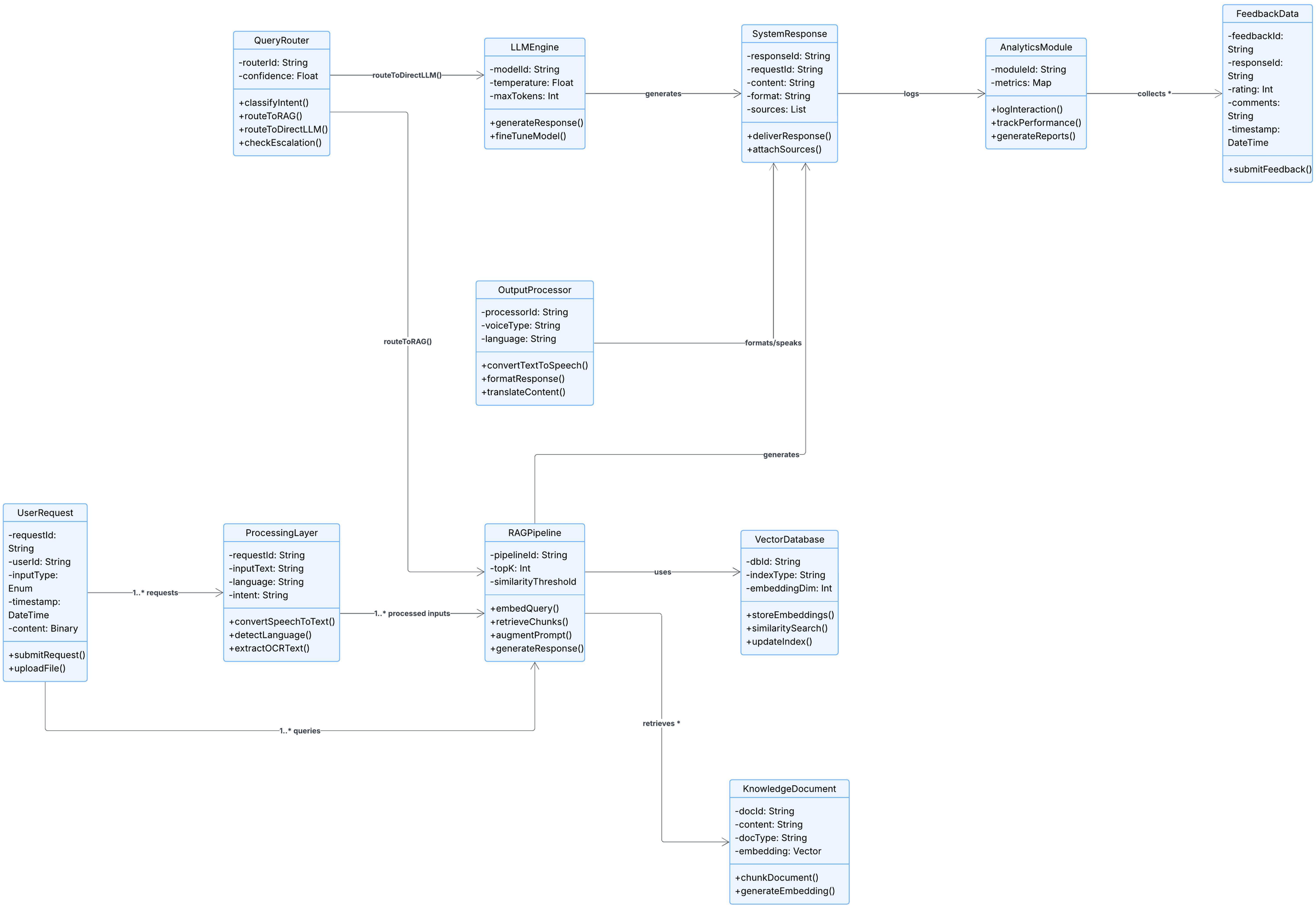
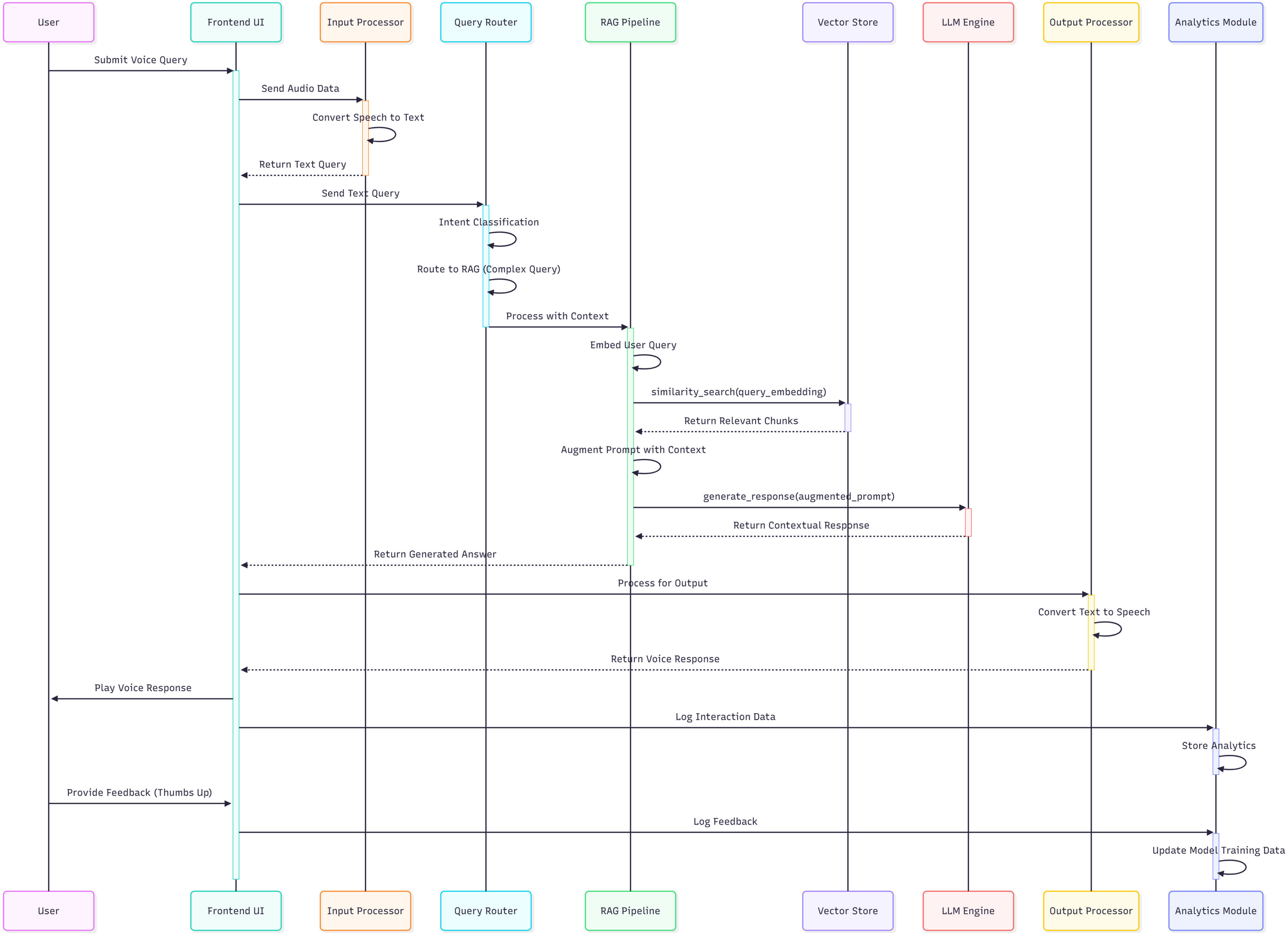
AI-Powered Query Processing Flowchart




User Initiates Request
Input Method
Text
Voice
Image
Convert Speech to Text
Extract Text via OCR
Validate Text Input
Detect Query Intent
Query Type
Simple FAQ
Complex/Contextual
Initiate RAG Pipeline
Embed User Query
Search Vector Database
Retrieve Relevant Chunks
Augment Prompt with Context
Generate Direct LLM Response
RAG LLM Generation
Output Format
Text Only
Voice
Format Text Response
Convert Text to Speech
Deliver Text Response
Deliver Voice Response
Collect User Feedback
Log Interaction & Analytics
Need Human Escalation?
No
Yes
Escalate to Human Agent
Process Complete

AI System Architecture Diagram




User
Frontend UI
Input Processor
Query Router
RAG Pipeline
Vector Store
LLM Engine
Output Processor
Analytics Module
Submit Voice Query
Send Audio Data
Convert Speech to Text
Return Text Query
Send Text Query
Intent Classification
Route to RAG (Complex Query)
Process with Context
Embed User Query
similarity_search(query_embedding)
Return Relevant Chunks
Augment Prompt with Context
generate_response(augmented_prompt)
Return Contextual Response
Return Generated Answer
Process for Output
Convert Text to Speech
Return Voice Response
Play Voice Response
Log Interaction Data
Store Analytics
Provide Feedback (Thumbs Up)
Log Feedback
Update Model Training Data
User
Frontend UI
Input Processor
Query Router
RAG Pipeline
Vector Store
LLM Engine
Output Processor
Analytics Module